

Apostol, Chapter - Exercise 19

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19. The Fibonacci sequence $1, 1, 2, 3, 5, 8, 13, 21, 34, \dots$ is defined by the recursion formula $a_{n+1} = a_n + a_{n-1}$, with $a_1 = a_2 = 1$. Prove that $(a_n, a_{n+1}) = 1$ for each n .

Let $n \in \mathbb{N}$. If $n = 1$, then $(a_{n+1}, a_n) = (1, 1) = 1$.

Now suppose $n \geq 2$. Obviously, $1|a_n$ and $1|a_{n+1}$.

Now let d be a common (positive) divisor of a_n and a_{n+1} . Since $a_{n-1} = a_{n+1} - a_n$, we have that $d|a_{n-1}$. Again, since $a_{n-2} = a_n - a_{n-1}$, we can conclude that $d|a_{n-2}$. Iterating this process we arrive at $d|a_2$. But $a_2 = 1$, so $d = 1$. Therefore, $(a_n, a_{n+1}) = 1$.